

2 The enzyme invertase breaks down sucrose into its two component monosaccharides.

(a) (i) Describe the structure of sucrose.

(2)

(ii) The effect of pH on the initial rate of the reaction catalysed by invertase was investigated at 25 °C.

State the independent variable in this investigation.

(1)

(iii) Suggest why the temperature was kept at 25 °C.

(2)

(iv) Name **one** non-biological (abiotic) variable, other than temperature, that should be kept constant in this investigation.

(1)

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(v) Describe how you would keep the variable you have named in (a)(iv) at a constant value.

(2)

(vi) Explain why the initial rate of this reaction was determined.

(2)

(vii) Suggest how the initial rate of this reaction could be determined.

(3)

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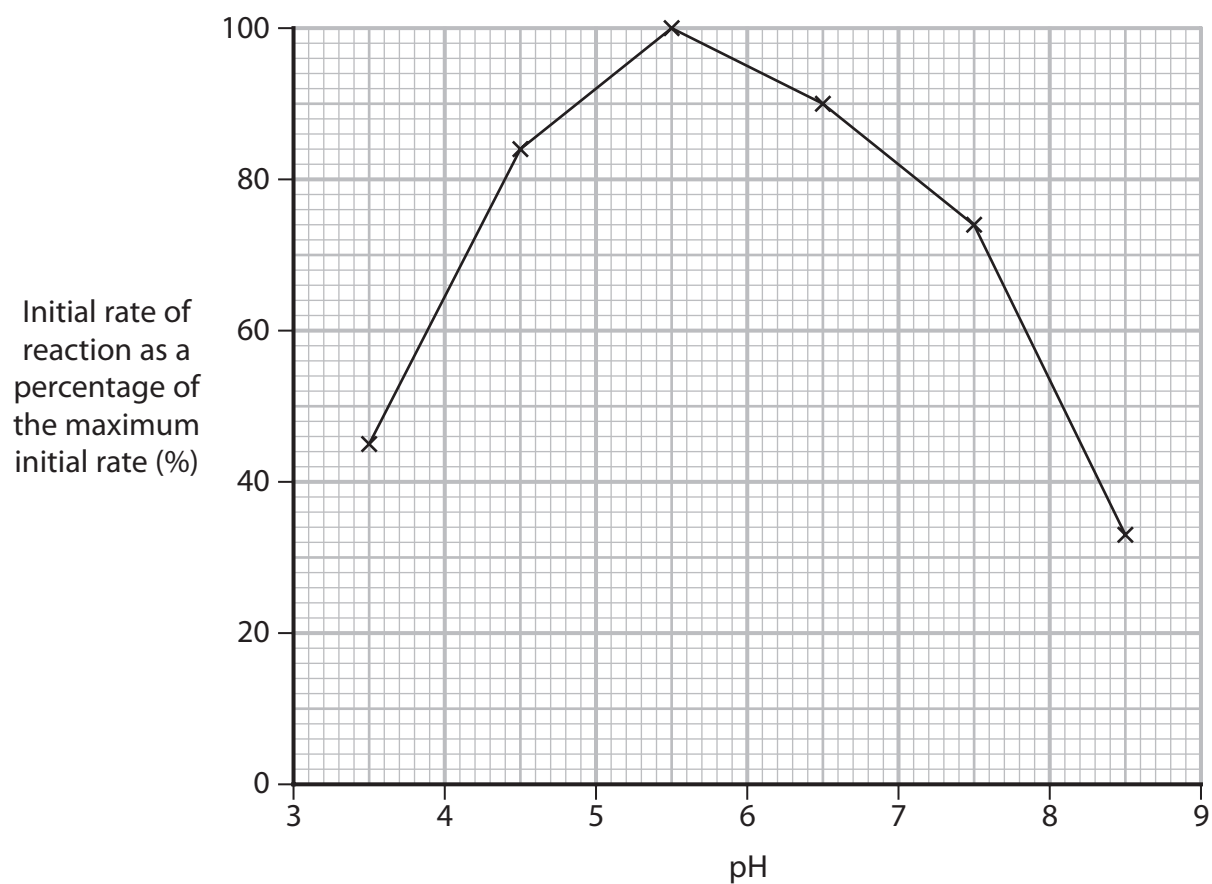
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P 6 2 4 6 0 A 0 9 2 0

(b) The graph shows the results of this investigation.



(i) Draw a suitable table to show these results.

(3)



(ii) Criticise the conclusion that 5.5 is the optimum pH for this enzyme.

(3)

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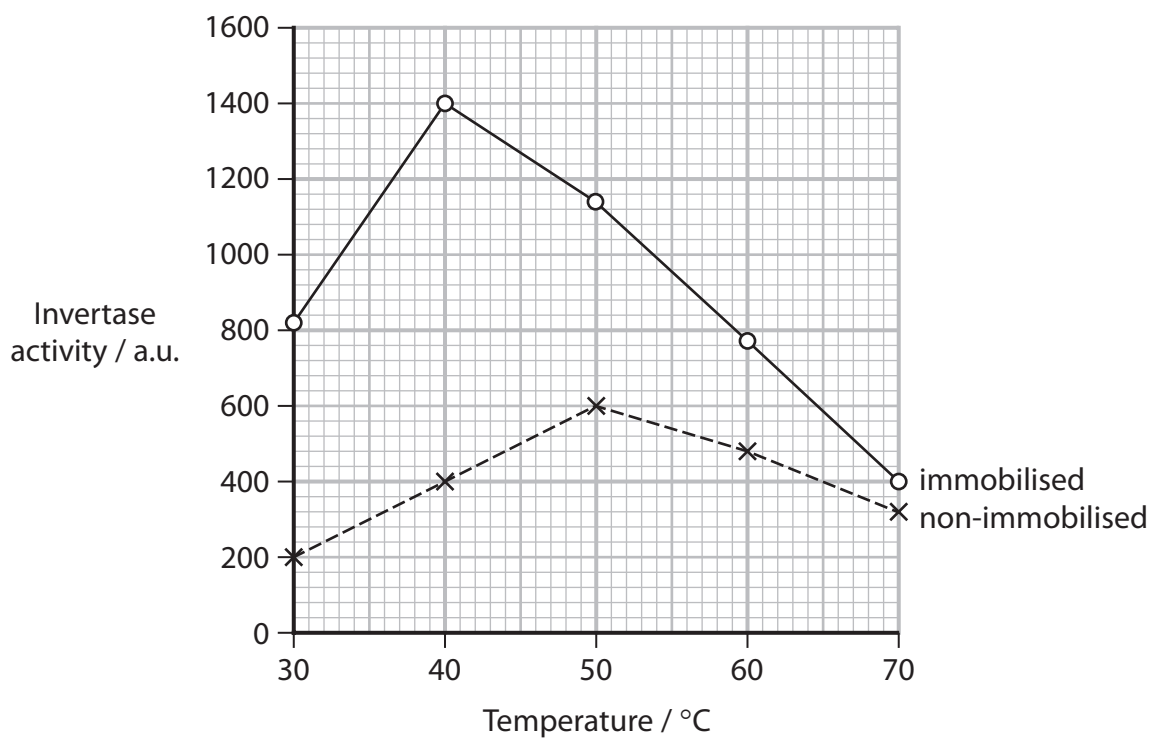


P 6 2 4 6 0 A 0 1 1 2 0

- (c) Invertase is used in the food industry. Scientists wanted to increase the efficiency of the enzyme to save time and money.

One way is to immobilise the enzyme by binding it to a surface. This stops the enzyme molecules from moving.

The effect of temperature on immobilised invertase and non-immobilised invertase was investigated. The results are shown in the graph.



Compare and contrast the effect of temperature on these two forms of invertase.

(3)

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(Total for Question 2 = 22 marks)

